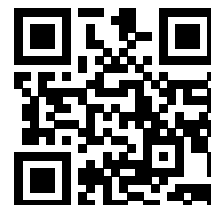


A many-designs study of 516 algorithms challenges the predictability of wisdom of the crowd forecast aggregation

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A many-designs study of 516 algorithms challenges the predictability of wisdom of the crowd forecast aggregation

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Data Availability Statement: We provide the preregistered study protocol and the data that support the findings of this study in an Open Science Framework (OSF) repository at <https://osf.io/kfnjd/>.

Code Availability Statement: The analysis code developed for this project is available via the Open Science Framework <https://osf.io/kfnjd/>. There are no restrictions on the use of the code we wrote for this project. The code is released under the MIT license, if not otherwise stated or licensed by other parties.

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Classification: Social Sciences; Psychological and Cognitive Sciences.

Keywords: Meta-science, Forecasting, Wisdom of the Crowd.

Abstract

Accurate forecasts are central to decision-making in many domains. Although aggregating independent judgments can improve forecast accuracy, a phenomenon known as “Wisdom of the Crowd,” methods for effectively combining these forecasts remain underexplored. In this preregistered many-designs study, 129 research teams independently submitted a total of 516 algorithms to aggregate forecasts over six months, across four domains: economics, politics, climate, and sports. By testing hundreds of independently developed aggregation algorithms under identical conditions, we established an empirical benchmark for forecast aggregation. Drawing on forecasts by 1,182 people, we evaluated algorithm accuracy and variability. The share of algorithms significantly outperforming the mean and median varied by domain, with the strongest gains over simple benchmarks in politics and climate. Algorithm performance was also more persistent over time in politics and climate than in economics and sports. When asked to predict algorithm performance, the participating researchers were overconfident about the success of their own submissions and of others. We also examined potential predictors of algorithm accuracy and variability. Although researchers expected various features to predict accuracy, no algorithmic or researcher feature consistently explained why algorithms varied in performance. The main exception was the use by algorithms of previously observed realized outcomes, typically to compute past forecast error or train a model, which predicted greater accuracy in politics and climate. The null findings contrast with prior evidence that certain weighting criteria can improve forecast aggregation, and illustrate the value of preregistered many-designs studies which report the full distribution of submitted approaches, revealing when previously supported approaches fail to generalize.

1 Introduction

Accurate prediction is essential for informed decision-making [1], and decades of research show that combining the forecasts of many individuals often yields more accurate results than relying on a single person [2, 3, 4]. This “Wisdom of the Crowd” (WoC) effect arises when a cognitively diverse group provides independent estimates, reducing individual errors and biases [5, 6, 7, 8]. Aggregating these individual responses produces more accurate estimates both for factual information with known answers [6] and for uncertain future events [9], rivaling the accuracy of domain experts [10].

However, research also shows that people fail to appreciate the benefits of averaging [11] and discount the judgments of others when forming their own estimates [12, 13]. Moreover, a single [14] or small set of top-performing individuals *can* outperform crowd aggregates [15]. These findings suggest that the benefits of aggregation are neither well understood nor guaranteed, raising the question of when and how it works best. Prior work has largely focused on narrow settings: examining single domains [16, 17, 18] or limited sets of aggregation rules [15, 19], and generally not examining the stability of aggregation performance [20, 21]. What remains missing is a comprehensive assessment of algorithmic aggregation methods in multiple socially relevant domains and over time. To fill this gap, we conducted a many-designs study to systematically evaluate the effectiveness of a wide range of aggregation algorithms across four forecasting domains by drawing on the contributions of a diverse set of expert researchers.

Given the importance of accurate forecasts, researchers from a range of fields such as psychology, statistics, political science, and economics have investigated the conditions under which collective forecasts are most accurate. Attempts to enhance the WoC can be broadly divided into two complementary strategies. *Elicitation* approaches aim to improve the way estimates are formed, often by fostering independence and diversity among individual contributors [22, 23] or by eliciting multiple perspectives before aggregation [24, 25], while *statistical* approaches aim to optimize the aggregation of individual estimates into a collective outcome, using various mathematical methods and weighting procedures [26, 27, 28, 29]. Statistical approaches are practical since they do not require

interventions during data collection, but their success depends on identifying the most effective aggregation algorithms. While a few studies have systematically compared a large number of statistical procedures [21, 30], none has, to the best of our knowledge, done so across a broad range of domains, and no study has used a many-designs approach to harness the creativity and expertise of many independent researchers for this purpose.

We address these gaps with a global, preregistered, crowdsourced forecasting competition, the *Wisdom of the Crowd: Crowd Analysis Project (WoCCAP)*, designed to evaluate a large set of aggregation methods for individual forecasts. In this many-designs study, independent research teams submitted algorithms for aggregating individual forecasts in four socially relevant domains: economics (asset prices), politics (presidential approval ratings), climate (hotter-than-average days), and sports (baseball outcomes). This novel approach to investigating the WoC allowed us to evaluate and compare many expert-generated aggregation algorithms in parallel, accelerating discovery and limiting the role of researcher discretion in selecting which methods to test. By reporting the full distribution of results, this approach also limits publication bias by including methods whose weak or null results might otherwise remain unpublished. Our analyses focused on algorithms’ forecasting accuracy and the persistence of their performance over time, on researcher expectations about algorithm success, on how algorithm characteristics shape researchers’ beliefs about accuracy, and on whether algorithm characteristics, researcher characteristics, and researcher beliefs predict actual performance.

In total, 129 research teams (RTs) from 29 countries—comprising 221 experts from diverse disciplines such as psychology, economics, data science, and the natural sciences—submitted valid aggregation algorithms. Each RT independently proposed algorithms to aggregate individual forecasts from US-based study participants in each of the four domains. All algorithms were submitted before any forecast data were collected and RTs could elect to use the same algorithm across all domains. In addition to descriptions of the forecasting task, the RTs received detailed descriptions of all participant-level measures, including demographics, confidence ratings, domain-specific knowledge, and past forecasting performance, as well as 29 other varied survey measures proposed and

voted on by the RTs, which could all be incorporated (e.g., as weighting criteria) into the algorithms. See Supplementary Table B1 for project terminology and Table B2 for a list of all measures in the forecast survey.

Forecast data was collected through Prolific in six consecutive monthly waves. We targeted 640 participants per month, encouraging repeated participation to establish a panel structure that allowed algorithms to take past forecasting performance into account. To that end, the first two waves were used to collect baseline forecasts but algorithms were only applied and evaluated starting in wave 3. In each wave, participants answered 16 forecasting questions: four in each domain, all with a one-month forecasting horizon, enabling a systematic comparison of 516 aggregation algorithms (129 per domain, 247 unique across domains).

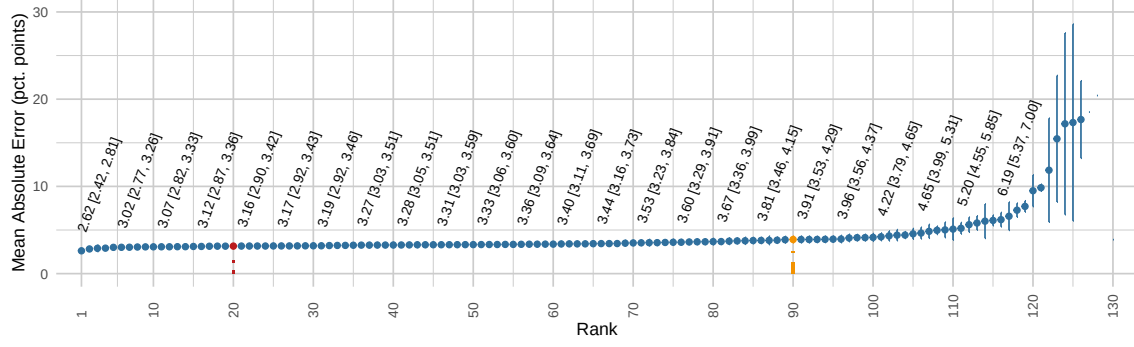
2 Results

We begin with preregistered analyses on the forecasting accuracy of the 516 algorithms in predicting real outcomes, as well as the persistence (stability) of their performance over time. We also present preregistered exploratory analyses on the extent to which RTs were overconfident in their expectations about algorithm performance. Next, we examine how various algorithm features influence peer beliefs about forecasting accuracy. Finally, we report the results of preregistered analyses testing whether algorithm and RT characteristics predict forecasting accuracy and variability. We also examine whether peer ratings of algorithm quality predict actual performance. Following [31], we adopt a significance threshold of $\alpha = 0.5\%$ to denote “statistically significant” results. Findings with a p -value between 0.5% and 5% are referred to as “suggestive evidence.”

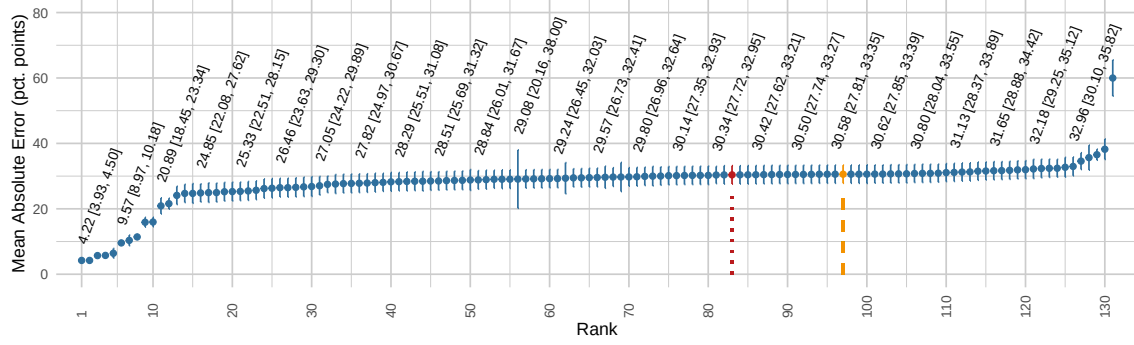
2.1 Performance, persistence and confidence in algorithm accuracy

We first examine the full distribution of algorithm performance. We observed notable differences, both within and across domains, in the accuracy with which the 516 submitted algorithms aggregated individual forecasts (Figure 1).

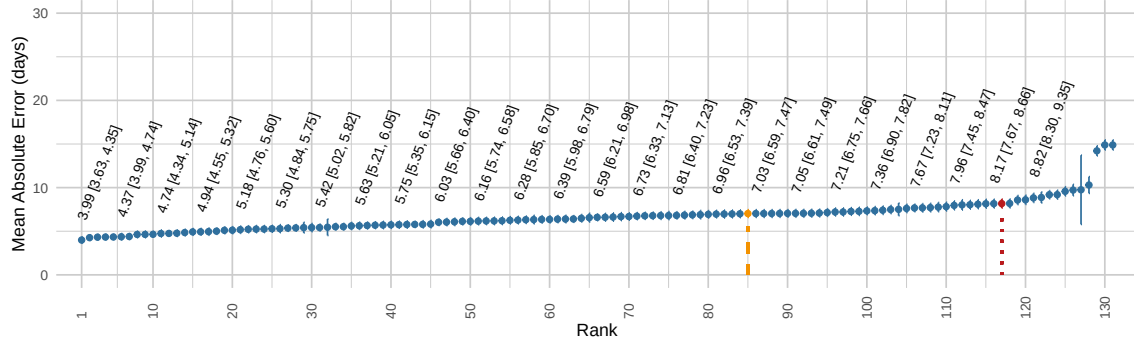
Algorithm accuracy, measured as the mean absolute error (MAE)—the average abso-



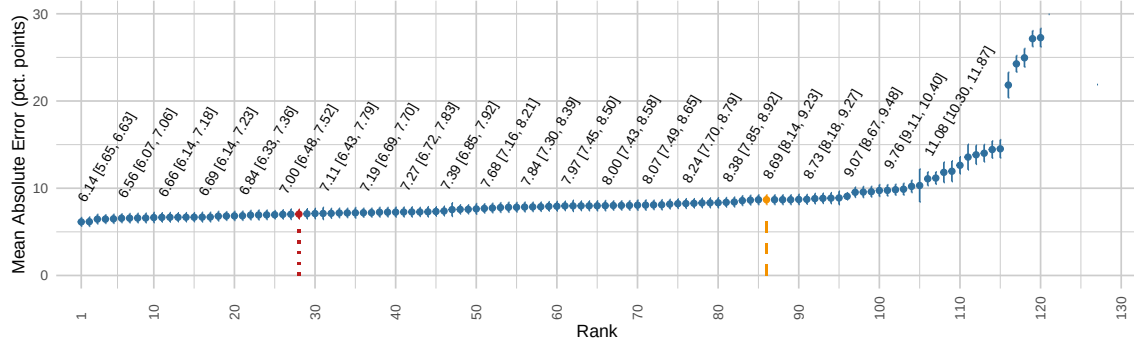
A: Economics



B: Politics



C: Climate



D: Sports

Figure 1: **Algorithm accuracy by domain.** Mean Absolute Errors (+/- standard error of the mean) for each algorithm and domain. Orange dashed line: *MEAN* algorithm; Red dotted line: *MEDIAN* algorithm. Some algorithms omitted because of high MAE.

lute difference between each aggregated forecast and the corresponding realized outcome (i.e., the true value)—varied between 1.33 and 1345.21 percentage points for economic outcomes, between 2.30 and 47.26 percentage points for political outcomes, between 2.78 and 14.25 days for climate outcomes, and between 1.50 and 54.83 percentage points for sports outcomes (see Supplementary Table B3 for realized outcome values). Similarly to many previous WoC studies [e.g., 3], we use the *MEAN* and *MEDIAN*, computed as the simple arithmetic mean and median of individual forecasts, as benchmark algorithms for comparison. We found that many algorithms suggested by researchers outperformed these benchmarks, especially for political and climate outcomes. Two-sample, two-sided Welch *t*-tests comparing the 10 best-performing algorithms to these benchmarks showed that all 10 were significantly more accurate than both the *MEAN* and the *MEDIAN* in these two domains. The pattern was less pronounced in the sports domain (*MEAN*: 6/10 sig., *MEDIAN*: 0/10 sig.) and virtually absent in the economics domain (*MEAN*: 1/10 sig., *MEDIAN*: 0/10 sig.); see Supplementary Tables B4 and B5.

To evaluate a basic WoC effect, we also compared the *MAE* of individual forecasts with that of *MEAN* and *MEDIAN* and found that these naïve benchmarks outperformed the average individual forecast in every domain (Table 1), demonstrating the benefit of simple aggregation. However, the best-performing algorithms were even more accurate than both *MEAN* and *MEDIAN*, highlighting the added value of more sophisticated aggregation strategies. The best-performing algorithm differed substantially across domains, but all used forecaster characteristics and realized outcomes from prior waves (e.g., to train their models or to estimate past forecasting performance) to improve forecast aggregation; full descriptions of the winning algorithms are provided in Supplementary Section G.

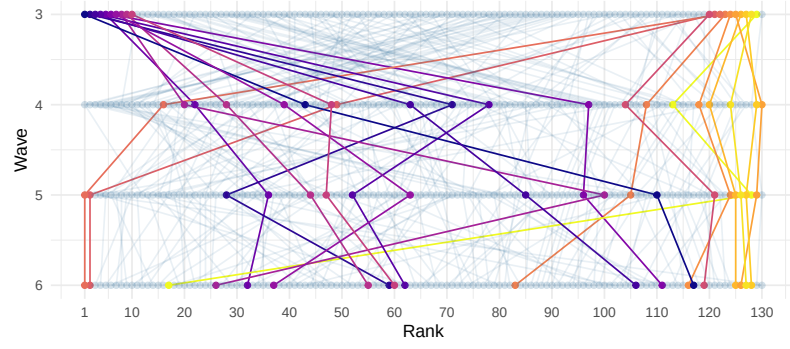
In a related exploratory analysis, we examined whether algorithm performance varied across domains. Using the Mean of Relative Errors (*MRE*; each algorithm’s absolute forecast error relative to the realized outcome value for the same question and wave), we ran two-sided Wilcoxon signed-rank tests to compare relative algorithm accuracy across domains and found that all pairwise differences between domains were statistically

significant ($p < 0.005, N = 126$). Relative error was lowest in sports ($MRE = 0.211$), indicating the greatest relative accuracy, followed by climate ($MRE = 0.506$), politics ($MRE = 0.862$), and economics ($MRE = 2.911$).

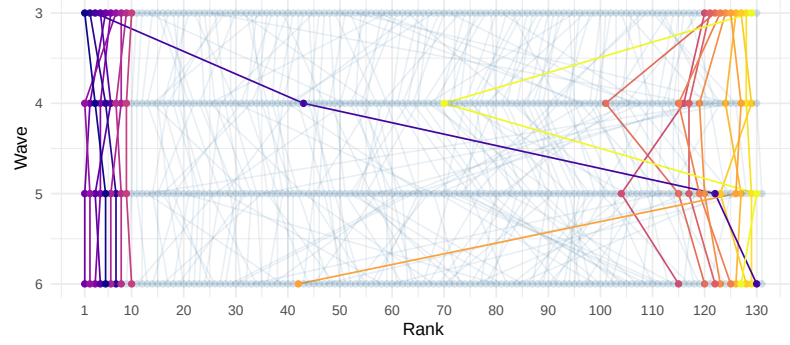
To examine the persistence of algorithm performance over time, we ranked them based on their performance in wave 3 (the first wave to which algorithms were applied), then tracked their ranks over subsequent months. Figure 2 tracks the best 10 and worst 10 algorithms in each domain throughout the four last waves: whereas there are substantial differences in the degree of performance persistence in the different domains, we observed rank persistence in some domains, particularly in the politics and climate domains, with both best- and worst-performing algorithms tending to maintain their relative ranks across multiple waves.

To quantify the persistence of algorithm performance, we calculated each algorithm’s rank in each wave and employed a linear mixed-effects model, which allows for random effects at the algorithm level and fixed effects on the wave level. We observed that there was no wave trend, but strong between-algorithm variation, meaning that performance differences were largely driven by the specific algorithm rather than the wave. The share of variability in algorithm rankings explained by which algorithm was used, rather than noise or wave-to-wave fluctuations, is represented by the Intraclass Correlation Coefficient (ICC). The higher the ICC , the more persistence there is in algorithm performance. Persistence as measured by the ICC was high in the climate (0.76) and politics (0.84) domains, and much lower in the sports (0.54) and economics (0.45) domains. This observation may reflect the variation of outcomes across domains, being highest in economics where, for instance, stock prices follow a random walk [32], affecting the persistence of algorithm performance.

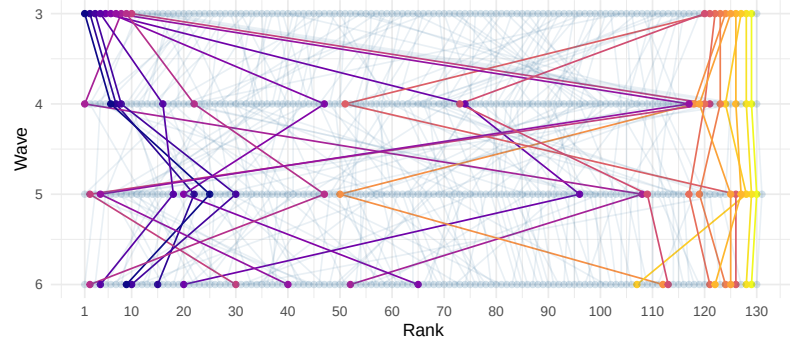
Next, we examine the confidence of RTs in algorithm accuracy (Figure 3), that is, their beliefs about how well their own and others’ algorithms would predict actual outcomes. When submitting their algorithms, RTs estimated the percentage of all algorithms that would produce aggregation results within ± 5 percentage points of the realized outcome for each domain. We ran exploratory regression analyses that compared these expecta-



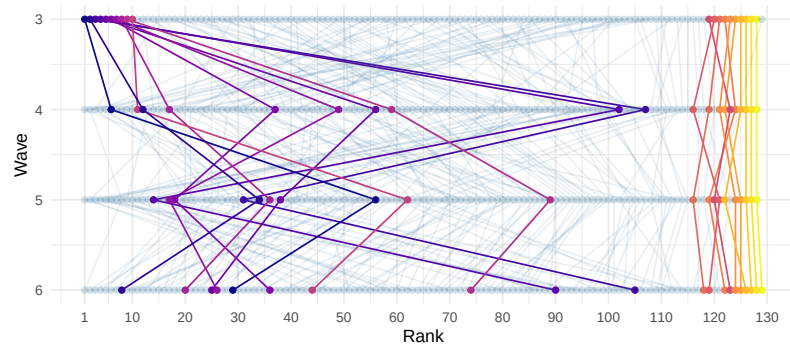
A: Economics



B: Politics



C: Climate



D: Sports

Figure 2: **Persistence of algorithm performance over time.** Algorithm persistence in ranks across all algorithms during the four main waves (months three to six) for the best 10 (dark purple hues) and worst 10 (light orange hues) algorithms of wave 3.

tions with the actual share of algorithms that achieved this accuracy. RTs estimated that 34.0 to 37.5% of algorithms (across domains) would fall within a 10 percentage point interval around the realized outcome, whereas only about 6% of algorithms did so. The difference was statistically significant in all domains ($p < 0.005$, two-sided tests of proportions, for individual test statistics see Supplementary Table B6).

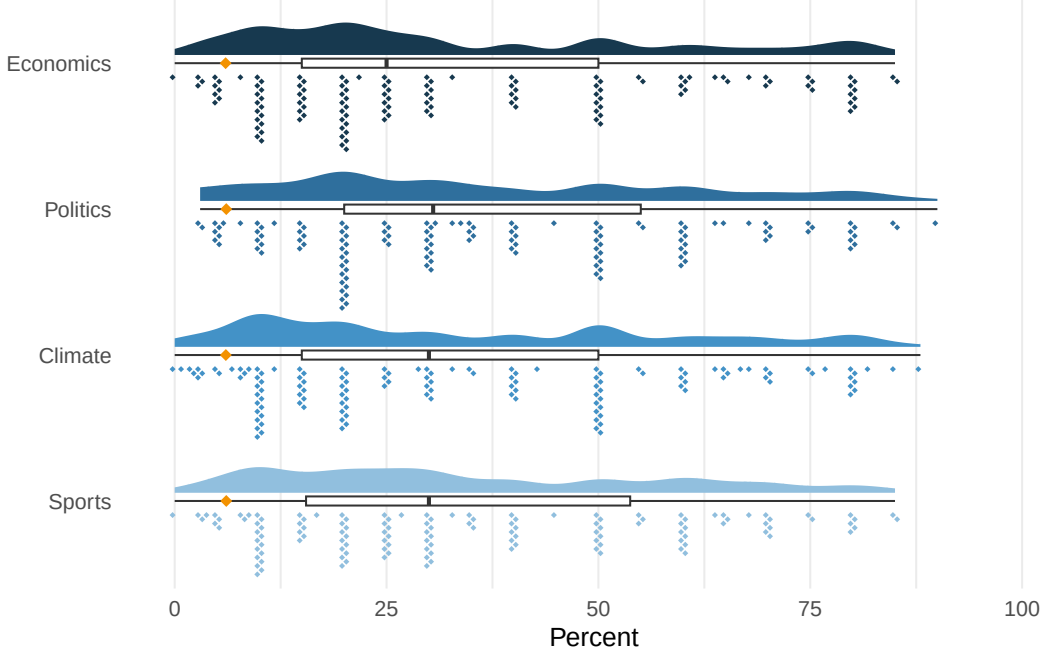


Figure 3: **Beliefs about algorithm performance.** Distribution of beliefs about the share of algorithm aggregation results falling within ± 5 percentage points of the realized outcome. Distribution, boxplot, and raw values by domain. Orange diamonds show the actual proportion of aggregation results that fell within ± 5 percentage points of the realized outcome. Also see Supplementary Table B6).

We also measured each RT’s confidence in their algorithms’ accuracy by asking them to predict the decile in which it would place in the final ranking. To determine actual performance, we ranked all RT algorithms within each domain based on their absolute error over the four-month period and assigned them to deciles (1 = best performance, 10 = worst). Comparing predicted and actual rankings using two-sided Wilcoxon signed-rank pairwise comparisons revealed consistent overconfidence: RTs expected better performance (lower deciles) than their algorithms achieved, in line with the better-than-average effect [33]. Actual deciles were significantly higher than predicted in all domains, significantly so in climate, economics, and sports, and suggestively so in politics (actual vs.

belief; climate ($N=129$): 5.47 vs. 4.41, $p = 0.003$; politics ($N=129$): 5.47 vs. 4.53, $p = 0.006$; economics ($N=128$): 5.44 vs. 4.38, $p = 0.002$; sports ($N=127$): 5.42 vs. 4.40, $p = 0.004$). Additionally, Spearman rank correlations between each RT’s overconfidence score (predicted minus actual decile) and their algorithm’s performance (rank), showed that those with the weakest performing algorithms (i.e., with higher ranks) were the most overconfident (climate ($N=129$): $\rho = 0.779$, $p < 0.005$; economics ($N=128$): $\rho = 0.774$, $p < 0.005$; politics ($N=129$): $\rho = 0.792$, $p < 0.005$; sports ($N=127$): $\rho = 0.790$, $p < 0.005$; p -values computed via the asymptotic t approximation).

2.2 Perceived drivers of algorithm accuracy

Next, we turn to the peer-rating stage, in which each algorithm was evaluated by 12 RTs to predict the rank decile in which it would fall, with lower deciles indicating better performance. We examined how various algorithm features influenced this peer-predicted performance with an OLS regression of the mean decile guess on variables indicating the degree to which each algorithm relied on different types of forecaster-provided information (Figure 4, panel A). These variables were based on ratings provided by the RTs during algorithm submission and included, in order of appearance in the survey: confidence, meta-forecasts (forecasters’ estimates of the average forecasts across all participants), performance on an unrelated estimation task (guessing the number of candies in a jar), domain knowledge (and experience), analytical skills, psychological measures, media consumption and literacy, economic variables, demographics and political views (see Supplementary Section E). These analyses yielded a mix of statistically significant and suggestive associations between algorithm features and peer-predicted performance. In the climate domain, RTs expected algorithms to perform worse if they relied on economic variables ($coeff = 14.235$, $SE = 6.483$, $p < 0.05$) and the candy estimation task ($coeff = 7.177$, $SE = 1.828$, $p < 0.005$) and better if they relied on demographics ($coeff = -22.317$, $SE = 7.993$, $p < 0.01$) and analytical skills ($coeff = -2.300$, $SE = 0.520$, $p < 0.005$). In politics, RTs expected algorithms to perform worse if they relied on analytical skills ($coeff = 1.666$, $SE = 0.739$, $p < 0.05$) and better if they relied on media

consumption and literacy ($coeff = -23.398$, $SE = 10.091$, $p < 0.05$). In sports, RTs expected algorithms to perform worse if they relied on confidence ($coeff = 13.342$, $SE = 4.983$, $p < 0.01$) and political orientation ($coeff = 20.004$, $SE = 9.088$, $p < 0.05$) and better if they relied on economic variables ($coeff = -18.421$, $SE = 8.953$, $p < 0.05$), media consumption and literacy ($coeff = -20.300$, $SE = 6.295$, $p < 0.005$), and meta-forecasts ($coeff = -8.787$, $SE = 3.727$; $p < 0.05$). In economics, algorithm features did not explain the variation in peer ratings, again underscoring the low level of predictability in this domain.

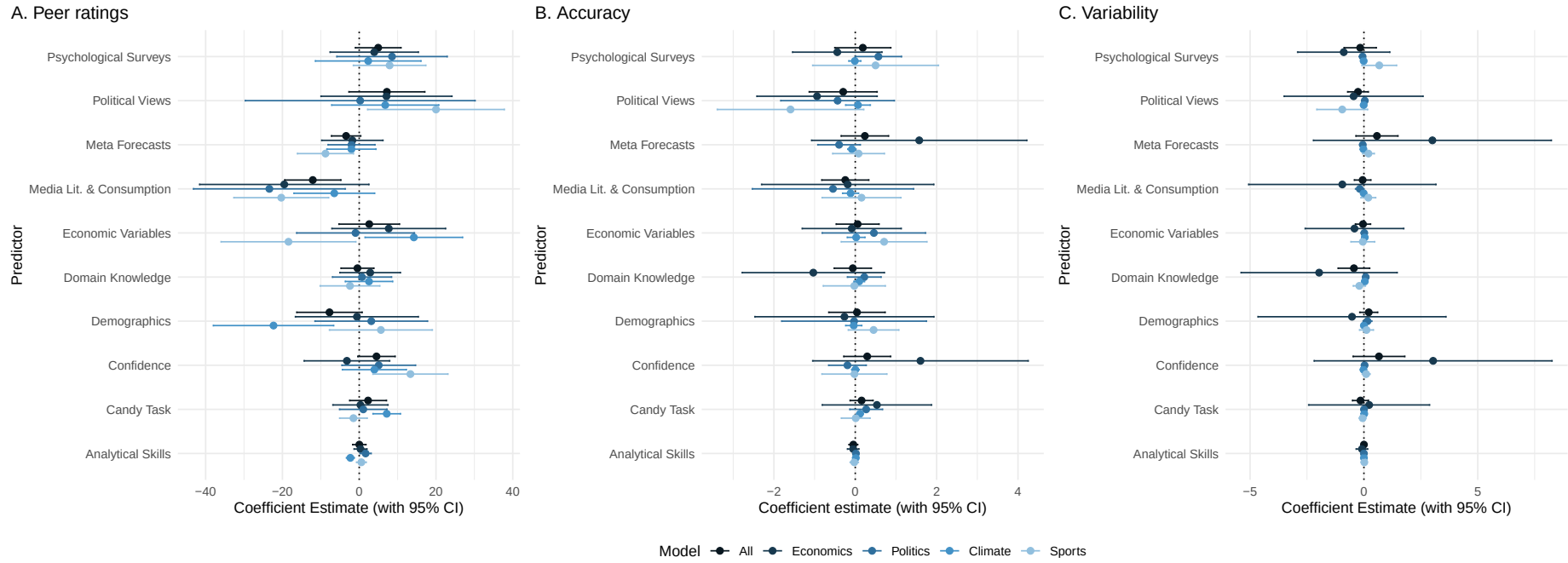


Figure 4: **Predictors of expected performance, algorithm accuracy, and variability.** Coefficient estimates from OLS regressions predicting algorithm outcomes from self-reported algorithm characteristics; error bars show 95% CIs (clustered). (A) RTs' expected rank decile for each algorithm; lower rank deciles indicate better expected performance, so negative coefficients indicate features associated with better expected performance. (B) Algorithm accuracy (z -transformed MAE); lower values indicate greater accuracy. (C) Algorithm variability (standard deviation of z -transformed MAE); lower values indicate more consistent performance.

2.3 Algorithm accuracy and variability

Finally, we turn to our preregistered analyses on the explanations of algorithm accuracy and variability. To test predictors of algorithm accuracy, we conducted OLS regressions with the z -transformed absolute error of each algorithm—the absolute difference between each algorithm’s aggregated forecast and the corresponding realized outcome—as the dependent variable for the entire dataset, and also for each domain separately. To test predictors of algorithm variability, we ran OLS regressions with the standard deviation of the z -transformed absolute errors, serving as the dependent variable. We applied clustered standard errors at the RT level in both regression types. As independent variables, we used the predefined full set of algorithm characteristics (see previous section), RT characteristics (team size, average self-reported expertise in forecasting / WoC research, position, doctorate degree), and RT predictions about how well others’ algorithms would perform (i.e., in which rank decile they would end up) from the algorithm peer-rating stage (details in the “Methods and methods” section). As preregistered, we also conducted additional coding of the algorithms and RTs (using AI-assisted methods) to capture supplementary (not pre-specified) characteristics, including each algorithm’s reliance on other weighting criteria (e.g., use of past forecast accuracy), the main aggregation mechanism (e.g., use of truncation, machine learning), and RT members’ domain expertise based on published research. This additional coding reduced the burden on RTs to provide extensive classifications and provided a more objective, standardized coding for certain characteristics, like publication-based expertise. To validate the AI-assisted classification, we manually reviewed a randomly selected subset of approximately 10% of the classifications and found high agreement with the AI-assisted coding.

Figure 4 (panel B) shows that the stated algorithm characteristics explain little of the variation in algorithm accuracy. Figure 4 (panel C) outlines results on algorithm variability and, again, the stated characteristics do not explain algorithm variability. No independent variable is statistically significant in any domain. Similarly, AI-coded algorithm characteristics largely did not predict algorithm accuracy and variability, with the exception of the use of past realized outcomes, which predicted greater accuracy in the cli-

mate and politics domain (Supplementary Tables C1, C2). This broad category included any algorithms that used past realized outcomes from the prior two waves to compute past forecast error, train the model applied to current crowd forecasts or, in a few cases, serve as direct inputs into the aggregation algorithm. Finally, none of the preregistered (Supplementary Tables C3, C4) or AI-coded RT characteristics (Supplementary Tables C5, C6) significantly predicted algorithm accuracy or variability.

We also preregistered an analysis of whether peer ratings predicted algorithm accuracy using OLS regression. We used the z -transformed mean absolute error as the dependent variable and the mean of peer ratings for a given algorithm as the independent variable, clustering standard errors at the RT level when pooling data across domains. We found a statistically significant and positive association only in the political domain, indicating that algorithms expected by peers to perform worse had higher actual error in that domain (all: $coeff = 0.010$; $SE = 0.011$; climate: $coeff = 0.001$; $SE = 0.001$; economics: $coeff = 0.038$; $SE = 0.039$; politics: $coeff = 0.013$; $SE = 0.004$; sports: $coeff = -0.004$; $SE = 0.007$). Finally, we investigated whether peer ratings explain algorithm variability (measured as the standard deviation of absolute error) and found no statistically significant effects (all: $coeff = 0.021$; $SE = 0.021$; climate: $coeff < 0.001$; $SE < 0.001$; economics: $coeff = 0.078$; $SE = 0.077$; politics: $coeff = 0.002$; $SE = 0.001$; sports: $coeff = -0.002$; $SE = 0.001$).

3 Discussion

Using a many-designs approach, we evaluated 516 algorithms submitted by 129 independent research teams to compare statistical procedures for aggregating human forecasts. Many algorithms outperformed the naïve MEAN and MEDIAN benchmarks, particularly in politics and climate, a noteworthy result given that simple averages often outperform more complex statistical approaches [9, 16]. Gains over these benchmarks (as well as the persistence of algorithm performance) were lower in economics and sports, possibly because greater inherent stochasticity of the outcomes [32, 34], or measurement-related noise limited the accuracy of forecasts and, in turn, forecast-aggregating algorithms. By con-

trast, political attitudes tend to be relatively stable over time [35], and climate outcomes exhibited predictable seasonal variation [36], potentially allowing aggregation algorithms to extract more predictive information from participant-level measures. Together these differences, both across and within domains, demonstrate the benefit of choosing the right aggregation strategy for the right outcome and suggest that differences in domain predictability may affect not only the accuracy of aggregated forecasts, but also the gains that aggregation algorithms can achieve over simple benchmark methods.

Yet, despite the variation in performance, we found few consistent predictors of algorithm accuracy or variability. These null results are surprising given previous evidence that weighting judgments by factors such as past forecast accuracy [28, 37], meta-predictions [20], knowledge and individual characteristics such as analytical skills [38, 39, 28] and confidence [38] can improve aggregate forecasts. Although many algorithms incorporated these features (Table 2), none predicted better aggregation accuracy. The main exception was the use of past realized outcomes, which predicted greater accuracy in climate and politics, suggesting that aggregation algorithms in these domains benefit from incorporating information about forecasters’ past accuracy or otherwise using past outcomes in the aggregation. (A similar, partial exception emerged for peer judgments: algorithms that peers expected to perform worse did, in fact, show higher error in the political domain.) At a descriptive level, exploratory comparisons between the top- and bottom-performing algorithms (Supplementary Tables D1 to D3) revealed several other modest contrasts, with top-performing algorithms more often using past forecasts and truncation rules (e.g., capping extreme forecasts), and bottom-performing algorithms more often using meta-forecasts and domain-knowledge.

Although these descriptive comparisons offer useful exploratory insights, the broader pattern is one of limited evidence for stable, systematic predictors of algorithmic accuracy. This absence of robust predictors echoes findings from large-scale many-analyst studies, in which extensive coding of researcher decisions has explained only a modest share of variation in reported outcomes [40]. This failure to identify systematic predictors of accuracy has several important implications. One possibility is that features identi-

fied as useful in prior research are context-dependent, improving accuracy only for some questions, domains or forecasting environments. Another is that these features improve accuracy only when combined with other algorithmic choices, suggesting that algorithm performance may be shaped by interactions among features rather than by any single feature alone. A third possibility is that the evidence of accuracy-enhancing features has been amplified by publication bias, or the preferential publication of significant results [41]. Publication bias may help explain why RTs expected certain theoretically promising features to improve performance even though those features did not reliably predict accuracy in our study. By evaluating a large set of features across the same forecasting tasks and reporting the full pattern of results, including null effects, our study provides more comprehensive evidence about whether algorithm features reliably improve performance. Importantly, given the breadth of the features tested, including those previously linked to accuracy, these null results are unlikely to reflect a failure to test the most plausible predictors.

Beyond misjudging which algorithm features would work, RTs also expected algorithms to perform about six times better than they ultimately did, suggesting that even expert researchers significantly underestimate the complexity of forecast aggregation. Also, despite knowing that they were competing against other self-selected and likely expert researchers, RTs overestimated their own algorithms' relative performance, consistent with previous evidence that even experts are prone to overplacement, or the belief that their judgments are superior to those of others [42].

This first application of a many-designs approach to WoC research has several benefits. First, it speeds up knowledge generation by enabling hundreds of researchers to work on the same problem simultaneously, facilitating direct comparisons across designs. Importantly, unlike prior many-designs studies which used independent designs to test if the findings converge on the same answer to an empirical question despite their heterogeneity [43, 44], here the designs themselves were also the object of the study. Second, by testing a broad range of independently developed algorithms, with only 22% implementing a previously established aggregation mechanism (Supplementary Table D1), the many-

designs approach reduced the role of researcher discretion. That means that the absence of consistent predictors of algorithm performance is less easily explained by a narrow or idiosyncratic choice of author-selected features. Third, the recruitment of multiple RTs permitted the examination of how researcher characteristics shape their algorithms. Finally, by publishing a detailed pre-analysis plan and committing to report the results of every submitted algorithm, this approach mitigated undesirable research practices such as *p*-hacking [45], forking [46], HARKing [47], and ignorance of null results [48].

The present study also comes with some limitations. First, we chose outcomes from four diverse domains which we could verify monthly, but these do not represent the full range of forecasting environments. Although we expect our findings to generalize to other similar quantitative forecasting tasks, future studies could explore whether they hold in other domains. Second, our use of a uniform quantitative forecast format ensured comparability between algorithms and domains, but limits the generalizability of the findings to other forecasting contexts, such as probabilistic forecasts of discrete events or events with longer time horizons [19, 49, 50]. Third, we chose a one-shot forecast elicitation format with no interaction, feedback, updating, or performance-based incentives because of evidence that independent, non-interactive forecasts produce the most accurate aggregate forecasts [6, 23], but also because this represents the most practical way to collect forecasts from large, randomly sampled groups. However, this minimalist setup may not generalize to more complex forecasting environments, such as prediction markets or crowdsourced tournaments, which often feature open participation [51], incentives for prediction performance [51], collaborative forecasting [52], and opportunities for forecast revision over time [19]. Finally, by rewarding the single best-performing algorithm per domain without penalizing poor performance, the competition structure may have encouraged teams to submit riskier, high-variance approaches to prioritize the chance of winning over consistent real-world accuracy [53].

In conclusion, this large-scale many-designs study provides both a comprehensive empirical benchmark for evaluating forecast aggregation algorithms and a reusable library of 247 algorithms in standardized format that future researchers can apply to new crowd

forecast data or extend to new domains. Despite the scale, diversity, and methodological rigor of our approach, we found no consistent predictors of algorithm accuracy and variability, an unexpected result given the prior research, the observed performance differences and some rank persistence. The mismatch between the performance predictors expected by RTs and actual predictors further highlights the complexity of designing forecast aggregation algorithms and underscores the challenges of aggregating human judgment and the need for continued cumulative research to uncover the latent structures that shape predictive success.

Materials and methods

The study protocol was preregistered on OSF and received ethical approval from the University of Innsbruck Institutional Review Board. The protocol and all relevant study materials are available on OSF: <https://osf.io/kfnjd/>. We, the project coordinators (PCs), invited the academic community to form research teams (RTs) and propose aggregation algorithms to combine forecasts in four domains: economics, politics, climate and sports. Once all algorithms had been submitted, we conducted a series of forecasting surveys over the course of six months to collect forecast data from participants recruited via Prolific. We calculated the accuracy of each algorithm and ranked them accordingly. In each domain, the RT submitting the most accurate algorithm won a monetary prize, but each member of every RT became a consortium co-author. We also collected RT- and algorithm-level characteristics. See Supplementary Table B1 for a summary of relevant project details.

Forecast surveys

To generate forecasts that can be aggregated by the algorithms, we recruited a US-based participant panel and administered forecast surveys to them in six monthly waves (from March to August 2024). Eligibility requirements included US residency, a past approval rate of at least 95% and completion of at least 100 studies. Participants were compensated at rates in line with Prolific’s recommended \$12/h (as of January 2024), with payments adjusted to the survey length. To discourage attrition, participants received a \$10 bonus for completing three sequential waves of data collection and a \$20 bonus for completing all six waves. We aimed to collect data from at least 640 participants per wave, providing a sufficiently large sample to generate multiple independent datasets for algorithm evaluation. Anticipating attrition, we started with many more participants in wave 1 and implemented refresher samples in waves 2, 3, and 4. The actual number of completed forecast surveys per wave were: 1,003 in wave 1, 820 in wave 2, 1,032 in wave 3, 1,055 in wave 4, 1,111 in wave 5, and 938 in wave 6. In total, 1,182 unique participants contributed data across the six waves.

In the first wave that each participant entered the study, they completed a baseline questionnaire assessing personal characteristics (see Supplementary Table B2 for an overview of survey measures and Supplementary Section E for exact survey questions). In each of the six waves, they answered the same 16 forecast questions (four per domain) about outcomes one month ahead. We chose four questions per domain to avoid algorithms being tailored to specific questions. For *politics*, forecasters were asked to predict by how many percentage points the presidential approval ratings (according to Gallup) would differ between (1) Democrats and Republicans, (2) White and non-Whites, (3) 18-29 year olds and 65+ year olds, and (4) college graduates and non-college graduates. For *economics*, forecasters predicted the one-month percentage change in the price of (1) WTI crude oil, (2) gold, (3) the USD/EUR exchange rate, and (4) the S&P500 index. For *sports*, forecasters predicted the percentage of games during the Major League Baseball season that would be won next month by (1) the Los Angeles Dodgers, (2) the New York Yankees, (3) the Chicago Cubs and (4) the Houston Astros. And for *climate*, forecasters predicted the number of days on which the average daily temperature would exceed the long-term average temperature recorded for that month over the past three decades at (1) Chicago O'Hare International Airport, IL, (2) JFK International Airport, NY, (3) Los Angeles International Airport, CA, and (4) Houston Intercontinental Airport, TX. For each participant, the order of the domains was randomized but remained fixed throughout the survey, and the four questions within each domain appeared sequentially on the same screen in randomized order, with each new question becoming visible only after the previous forecast and confidence rating had been entered.

In addition to a limited set of core measures, such as demographic questions, we asked RTs to propose additional survey items. These could take the form of individual questions, validated scales, or short economic games, provided they adhered to key constraints: no deception, no physical risk, voluntary participation, no participant interaction, and no monetary incentives. All proposed measures were required to include clear instructions, specify the estimated completion time, indicate whether they would appear only in the first or in all waves, and include a list of target variables.

The initial baseline survey (i.e., first wave of participation), including all core and additional measures, was designed to take approximately 45 minutes. Because most of the measures were collected only once, subsequent waves took approximately 10 minutes. To accommodate this time constraint while ensuring a broad representation of RT contributions, RTs could review and endorse the proposed measures submitted by others. Endorsements were not publicly displayed, but were used by the PCs when selecting which measures to include. After excluding duplicate proposals and questionnaires already included as sub-questionnaires in other measures, we implemented every other proposed measure in a pilot study with Prolific respondents ($N = 50$) to determine how long each took to complete. We prioritized the measures based on the ratio of median completion time to endorsement count. Based on these criteria, 29 of the 44 proposed measures were included in the forecast surveys (for a total of 35 unique measures when including core measures) and made available to all RTs to use as algorithm criteria. See Supplementary Table B2 for a list of all included measures and Section E for descriptive statistics.

Crowdsourced algorithms

To crowdsource aggregation algorithms, we issued an open call for participation (mainly via professional society newsletters, but also social media), inviting researchers to form small teams of one or two members. RTs were eligible for participation if at least one member had a doctorate degree or an active affiliation with a research institution. RTs were required to provide member details, including name, email, field of research, affiliation, position, doctoral status, and year (if applicable), and optional identifiers such as ORCID and academic web profiles. Teams also reported their expertise in forecasting, prediction markets, aggregation algorithms, and WoC research, and disclosed any relevant declarations or competing interests. Details about the participating RTs are available in Supplementary Section F. All RT members who participated in every phase of the project were included as consortium co-authors on the present paper. In addition, a WoCCAP Award of €2,500 was awarded to the RT with the most accurate algorithm in each of the four domains, for a total prize pool of €10,000. Details about the competition

winners and results are available in Supplementary Section G.

Teams received codebooks for forecast survey datasets and a demo dataset with randomly generated data to help them develop and test their algorithms. Each RT was required to submit four aggregation algorithms—one for each forecast domain, with each algorithm using only data from its respective domain. The submitted algorithms had to (1) use the forecast data, (2) not simply take a mean or a median, and (3) be implemented in *R* or *Python* code. All algorithms were first tested on the demo data via an automated system to verify for errors. Once finalized, each algorithm was submitted along with a plain-language description and a classification form specifying which measures it relied on. Algorithms were included in our analyses only if they did not produce any errors, but we retained algorithms that occasionally failed to produce an outcome for certain datasets, for example, when exclusion criteria filtered out all individual forecasts.

RTs also rated the extent to which their algorithm relied on different measures from the survey using an unmarked slider ranging from “not at all” to “extremely.” The categories were: (1) confidence, (2) meta-forecasts (i.e., beliefs about others’ forecasts), (3) performance on an unrelated estimation task, (4) domain knowledge, experience, and preferences, (5) mathematics, statistics, finance, and analytical skills, (6) psychological measures, (7) media consumption and literacy, (8) economic variables, (9) demographics and (10) political views. Participants first indicated which characteristics their algorithm relied on, and, for the selected ones, indicated how much weight the algorithm put on the respective characteristic using the unmarked slider, internally coded as 0 to 100. We transformed these weights into relative weights by dividing each weight by the sum of the weights of all used characteristics. This accounts for the fact that people may select different numbers of characteristics and use the scales differently.

Forecast datasets

Each algorithm was applied to a series of forecast datasets and each dataset consisted of a randomly sampled subset of 80 individual forecasters from a given wave, yielding 80 forecasts for a specific forecast question. Subdividing the participants from each wave

into multiple such subsets allowed us to generate a larger number of independent datasets for statistically comparing algorithm performance. In addition to current forecasts, the dataset included the two most recent forecasts for each participant (from the two prior waves of data collection), the corresponding two realized outcomes, forecast confidence ratings, meta-forecasts (i.e., estimates of what other participants predicted), and personal characteristics. In order to include forecasts from the prior two waves in each dataset, algorithms were applied only to the last four of the six waves. We planned to apply each algorithm to 128 such datasets ($4 \text{ questions} \times 4 \text{ waves} \times 8 \text{ datasets of participants}$) but ended up with 108 observations per algorithm due to a small data loss in wave 1 (approximately half of the page time data from wave 1 was not downloaded). We oversampled in subsequent waves to compensate, but this resulted in fewer datasets in wave 1 ($N = 16$) and wave 2 ($N = 28$) than planned. Waves 3 and 4 reached the intended number of datasets ($N = 32$ each). As a result, each algorithm was applied to 108 datasets instead of the planned 128. This shortfall does not affect the integrity of the competition, as the same datasets were used for all algorithms.

Peer ratings

Each RT was also required to evaluate a subset of algorithms (3 per domain) submitted by other teams and to predict their relative accuracy. They were provided with the plain language description of the algorithm as well as the source code and assigned algorithms in the same programming language as their own. Specifically, each RT was asked to estimate how close each algorithm’s aggregated forecast would be to the realized outcomes (measured by absolute deviation) by guessing the decile in which the algorithm would rank in the final accuracy distribution. For each domain, we randomly selected one of the correct guesses (i.e., for which the RT correctly guessed the accuracy decile) and awarded that RT 100 EUR.

Details on study design, data quality and data analysis

Study design

A noteworthy feature of the study design is the use of structurally similar forecasts within each domain, a deliberately different approach from highly heterogeneous prediction tournaments [50], which include many unrelated questions. Whereas such tournaments aim to evaluate how well individuals make forecasts across a wide range of topics, our study aims to evaluate the algorithms used to aggregate those forecasts. To that end, the main purpose of the four questions within each domain is to allow us to apply each algorithm to multiple related forecasts, thereby increasing the number of observations per algorithm without relying on a single, potentially idiosyncratic outcome. In addition to being methodologically necessary, our streamlined approach is in line with established traditions in the WoC literature. Economic research on forecast combination often compares different aggregation schemes by applying them to related economic indicators such as GDP growth and inflation [16] or the same indicator across multiple countries [17] and psychology research often uses highly related judgment tasks [24, 54]. Although, as in these studies, we did not seek to maximize the variety of forecast outcomes within domains, the inclusion of four distinct domains allows us to evaluate algorithms across substantially different forecasting contexts. Within those contexts, the selected questions, while structurally homogeneous, reflect important, widely tracked outcomes that capture common forecasting problems and provide a reasonable representation of typical forecast tasks.

For the forecasts, we elicited direct point estimates of quantitative outcomes (e.g., what the inflation rate would be next month) rather than probabilistic forecasts of discrete or threshold events (e.g., the probability that the inflation rate would exceed a certain value). Point estimates made it straightforward to evaluate each algorithm, because each produced an aggregated estimate that could be directly compared with the realized value. While probabilistic forecasts are common in forecasting tournaments [49], point estimates are consistent with the WoC literature on aggregating numeric judgments or forecasts [2, 15, 55], as well as with forecast-combination research [9, 16].

Data analysis

Data-quality checks for participant responses are reported in the Supplementary Information. Attention-check performance (see Supplementary Table E72), comprehension checks (Supplementary Table E73), reCAPTCHA scores (Supplementary Table E83), and multi-item scale reliabilities (Supplementary Table E84) all indicated acceptable data quality. Although this research is not mainly concerned with the quality of the forecasts but with the quality of the aggregation algorithms, it is important to note that the forecast data contained meaningful signals and were informative enough for basic aggregation to work, as demonstrated by the fact that simple aggregation rules not only outperformed the average individual forecaster but also outperformed many submitted algorithms. And, importantly, because every algorithm was evaluated on the same datasets, any remaining noise in the forecasts was held constant across algorithms.

The analyses and visualizations included only the aggregation results if they fell within domain-specific bounds reflecting the range of meaningful values. For climate, aggregate results had to be between 0 and 31 days (inclusive). For economics, they had to be no smaller than -100 , as outcomes were expressed in percentage change. For politics and sports, they had to fall between 0 and 100 percentage points (inclusive). To manage the influence of extreme values, particularly in the economics domain, we truncated visualizations at the 95th percentile of absolute error. However, these statistical outliers were retained in all inferential analyses unless they violated the plausibility criteria described above.

Unless otherwise specified, algorithm accuracy was assessed using the z -transformed absolute error (zAE), computed within each forecast dataset. The AE refers to the absolute error of an algorithm's aggregated forecast and the z -transformation standardizes these errors across algorithms for a given question, domain, and wave. This standardization enables comparisons of algorithm performance across domains, questions, and waves.

We conducted two levels of analysis, as outlined in the pre-analysis plan. For our primary analyses (comparing algorithm accuracy and performance over time) the unit of

analysis was the application of a single algorithm to a specific forecast dataset, defined by the domain, forecast question, wave, and subset of forecasters. This yielded up to 108 observations per algorithm. To test whether the algorithm features predicted accuracy or variability, we aggregated the performance across these datasets, computing the mean or standard deviation of zAE within each domain for each algorithm. These aggregate values served as dependent variables in our regression analyses on algorithm and team characteristics, as well as peer evaluations.

To assess potential multicollinearity, we examined pairwise correlations and variance inflation factors (VIFs) for all regression models. Most correlations were low to moderate, and most VIFs ranged between 1 and 5 (see Supplementary Section H). Among the 10 self-reported algorithm characteristics, three VIFs exceeded 10 in the economics domain and one exceeded 10 in the sports domain (Supplementary Tables H13, H16, and H34). Among the 11 AI-coded team-expertise measures, two exceeded 10 in each domain and in the pooled analysis (Supplementary Table I30). Because the same variables did not reliably predict performance across domains or analyses, these isolated cases of multicollinearity do not alter the main conclusion that algorithm and team characteristics showed little consistent predictive value.

Our sample size choices were guided by both prior research and practical considerations. At the dataset level, we used 80 forecasters to ensure reliable aggregate estimates, and prior work on crowd size effects shows that most of the WoC benefit is relatively small group sizes [56]. We did not conduct an ex-ante power or sensitivity analysis to choose the more relevant sample size for our analyses (108 datasets per algorithm) because no prior work provided plausible effect-size estimates for algorithm performance in this context, and the multi-level, multi-metric nature of our analyses (spanning hundreds of algorithms and multiple domains) complicates such calculations. Nevertheless, this design produces a sample size that is comparable to, and in some cases larger than, those used in related aggregation studies.

Our reported standard errors for algorithm performance capture the variability in the aggregated forecast errors across datasets but do not account for the additional sampling

variance that arises because each algorithm’s forecast is based on a single random sample of 80 forecasters. As a result, our standard errors likely underestimate the true variability in algorithm performance, but this does not bias comparisons between algorithms, which are based on relative performance rankings.

The forecast datasets included realized outcomes from the two prior waves so that RTs could compute past forecaster performance. The winning RTs were determined based on the lowest mean absolute error across all waves. One RT achieved the top score in two domains, but was disqualified because its submissions did not rely on the individual forecast data provided for aggregation and instead used only realized outcomes from the previous two months as inputs, which violated the competition rules. We therefore excluded this team’s submission from all reported results. Several included algorithms, including some high-performing ones, combined current crowd forecasts with prior realized outcomes. This was technically allowed when prior outcomes were used together with participant forecasts, and suggests that, for some short-horizon forecasting tasks, combining crowd forecasts with recent outcome information may be effective.

We made several deviations from our pre-analysis plan. For example, we report analyses comparing the top 10 performing algorithms in each domain to these benchmarks, but, in our pre-analysis plan, we specified that we would compare the top 10% of algorithms. All deviations can be found in the Supplementary Sections H to J.

AI-assisted classification

We used large language models (LLMs) to gain additional insights into the inner workings of the algorithms submitted by the RTs. Specifically, we used ChatGPT 4o (model gpt-4o-2024-08-06) through OpenAI’s API. The system prompt described the purpose of the code and instructed the model to classify whether the algorithm made use of certain variables or used specific analysis techniques. User prompts contained the description of the algorithm provided by the RTs and the relevant algorithm code. Each algorithm was processed in a separate request. We used the structured output format option and enforced binary responses. For more information, refer to Supplementary Section I.

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Table 1: Mean absolute error of individual forecasts, of the mean and median algorithms, and of the best algorithm by domain

Type	Economics	Politics	Climate	Sports
Mean of individual forecasts	6.33	30.83	9.53	16.68
Mean algorithm	3.91	30.58	7.03	8.69
Median algorithm	3.16	30.38	8.17	7.03
Best algorithm	2.62	4.22	3.99	6.14

Table 2: Prevalence of algorithm characteristics by domain

Algorithm characteristic	Economics	Politics	Climate	Sports	Overall
Variables used by the algorithm					
Confidence weighting	0.46	0.43	0.45	0.43	0.44
Domain knowledge weighting	0.36	0.33	0.38	0.39	0.37
Domain expert weighting	0.23	0.22	0.25	0.25	0.24
Meta-forecasts	0.26	0.22	0.22	0.21	0.23
Participant characteristics	0.33	0.41	0.35	0.28	0.34
Past forecast errors	0.55	0.54	0.57	0.56	0.56
Past forecasts	0.46	0.44	0.45	0.46	0.45
Past realized outcomes	0.65	0.64	0.65	0.65	0.65
References prior literature	0.18	0.19	0.17	0.19	0.18
Algorithmic methods					
Mean/median aggregation	0.78	0.74	0.75	0.76	0.76
Truncation or winsorizing	0.29	0.29	0.31	0.29	0.29
Existing mechanism included	0.22	0.24	0.20	0.21	0.22
Regression-based weighting	0.19	0.20	0.19	0.19	0.19
Machine learning approach	0.13	0.12	0.15	0.12	0.13

Note. Values represent the proportion of submitted algorithms classified as including each feature (0 = absent, 1 = present). Classifications were generated using AI-assisted coding of the algorithm descriptions and code (see Supplementary Section I.5). “Past realized outcomes” indicates use of past realized outcome values from the prior 2 waves. Manual review indicated that only four algorithms in each domain used these outcomes directly, rather than to calculate past forecast errors. One additional algorithm relied exclusively on past realized outcomes and was excluded from all analyses.

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A many-designs study of 516 algorithms challenges the predictability of wisdom of the crowd forecast aggregation

Abstract

Accurate forecasts are central to decision-making in many domains. Although aggregating independent judgments can improve forecast accuracy, a phenomenon known as “Wisdom of the Crowd,” methods for effectively combining these forecasts remain underexplored. In this preregistered many-designs study, 129 research teams independently submitted a total of 516 algorithms to aggregate forecasts over six months, across four domains: economics, politics, climate, and sports. By testing hundreds of independently developed aggregation algorithms under identical conditions, we established an empirical benchmark for forecast aggregation. Drawing on forecasts by 1,182 people, we evaluated algorithm accuracy and variability. The share of algorithms significantly outperforming the mean and median varied by domain, with the strongest gains over simple benchmarks in politics and climate. Algorithm performance was also more persistent over time in politics and climate than in economics and sports. When asked to predict algorithm performance, the participating researchers were overconfident about the success of their own submissions and of others. We also examined potential predictors of algorithm accuracy and variability. Although researchers expected various features to predict accuracy, no algorithmic or researcher feature consistently explained why algorithms varied in performance. The main exception was the use by algorithms of previously observed realized outcomes, typically to compute past forecast error or train a model, which predicted greater accuracy in politics and climate. The null findings contrast with prior evidence that certain weighting criteria can improve forecast aggregation, and illustrate the value of preregistered many-designs studies which report the full distribution of submitted approaches, revealing when previously supported approaches fail to generalize.

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